

Performance Analysis

Business objective

This project evaluates marketplace growth, product and seller performance, delivery reliability, customer reviews, and geographic risk using SQL and Power BI. The objective is to identify operational priorities that could improve customer experience and marketplace performance.

Key findings

- Delivered-order GMV grew strongly during 2017. Monthly performance fluctuated during the available January–August 2018 period.
- The late-delivery rate increased from 6.4% in 2017 to 9.2% in 2018.
- The overall average review score was 4.05/5 across the selected dataset. Late-delivered orders averaged 2.53/5 in 2018. These figures refer to different time scopes.
- Health and Beauty and Watches and Gifts were the leading categories by delivered-order GMV across the full analysis period.
- Delivery risk varied across states: some states had high late-delivery rates, while larger markets accounted for more late orders in absolute terms.

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Recommendations

- Prioritize delivery investigations using both late-order volume and late-delivery rate. High-volume markets and high-rate smaller markets require different interventions.
- Investigate seller dispatch times and delivery durations on high-risk routes before recommending operational changes.
- Examine monthly changes in order volume and order value separately. Compare equivalent periods and account for different month lengths before attributing changes to seasonality

Tools

MySQL Workbench | SQL | Power BI | GitHub

* GMV - gross merchandise value

2016 | 2017 | 2018

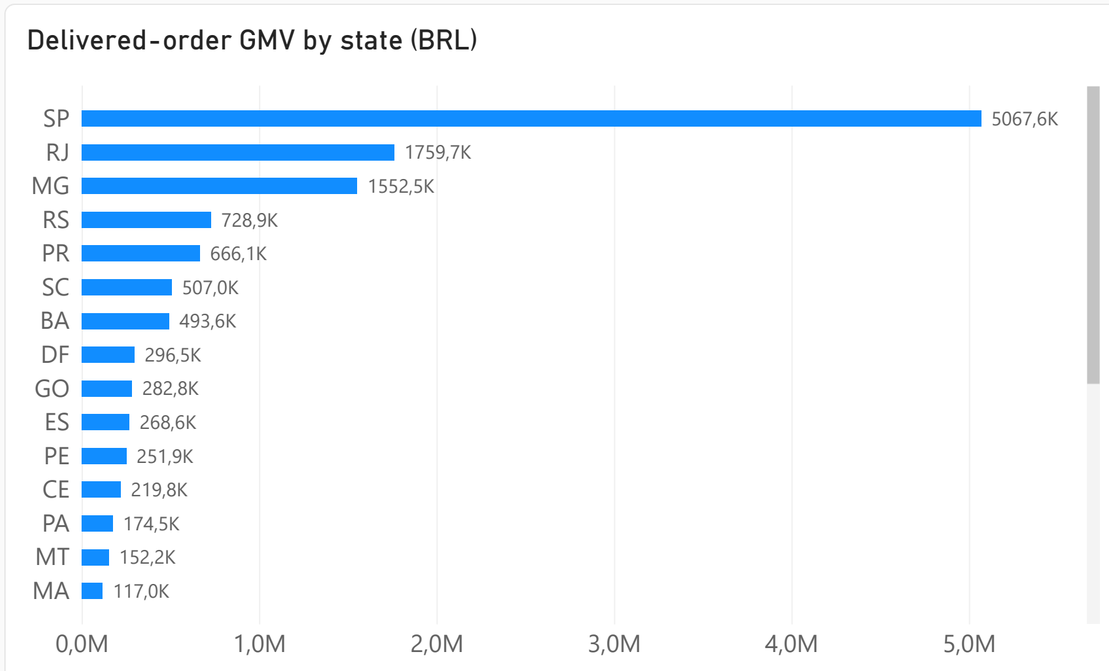
Marketplace overview:

KPIs, monthly performance, and state-level GMV

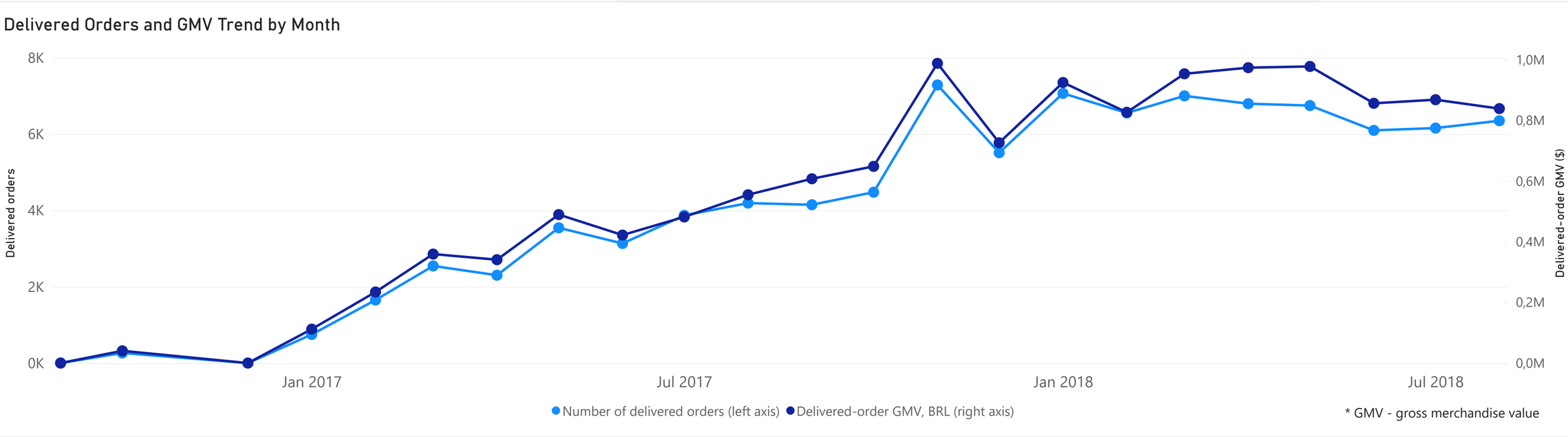
(for delivered orders)

Selected period: 2016, 2017, 2018

96,5K	13,22M	4,13
Delivered orders	Delivered GMV	Average review score



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BRAZILIAN E-COMMERCE



2016 | 2017 | 2018

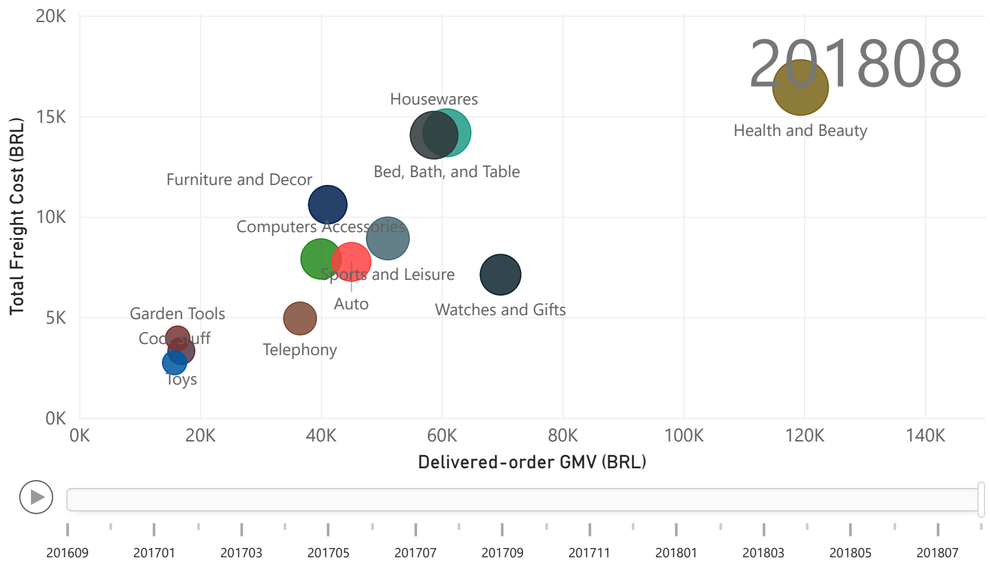
Marketplace Product and Seller Performance

Delivered-order GMV, item revenue and freight cost by product category and seller

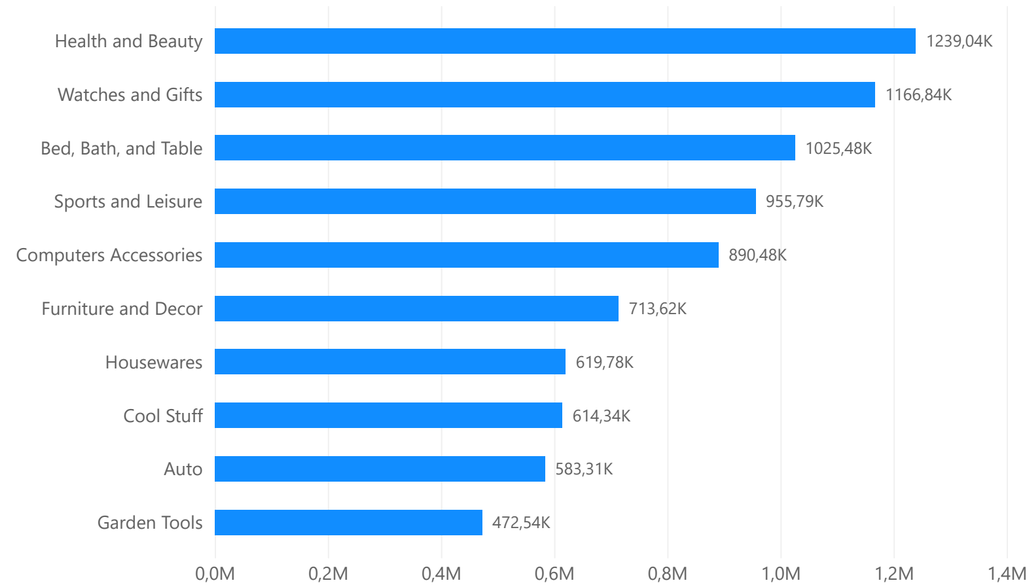
Selected period: 2016, 2017, 2018

100,0%
Category Share of GMV

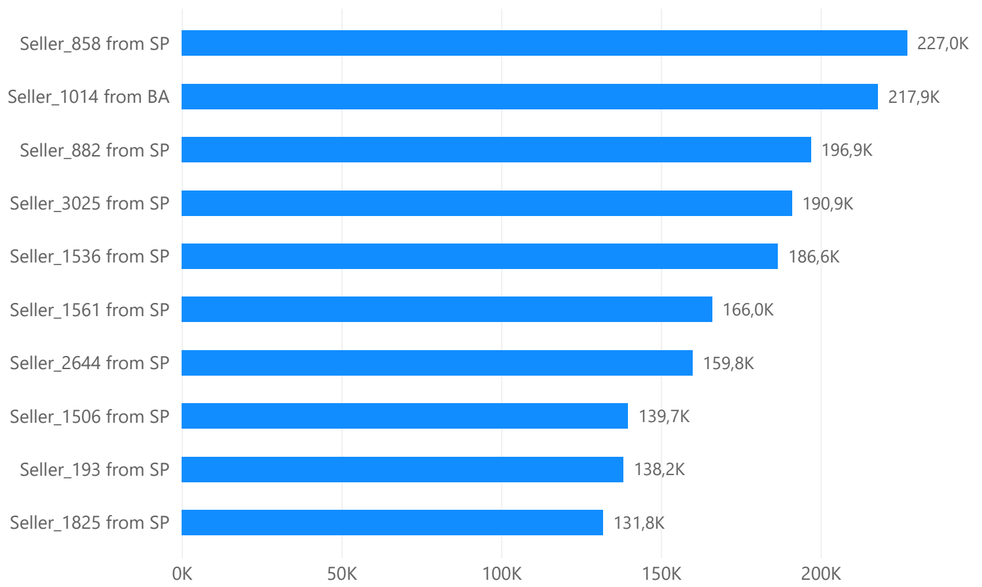
Top 12 product categories by freight cost: GMV and order volume



Top 10 categories by delivered-order GMV (BRL)



Top 10 sellers by delivered-order GMV (BRL)



* GMV - gross merchandise value

2016 | 2017 | 2018

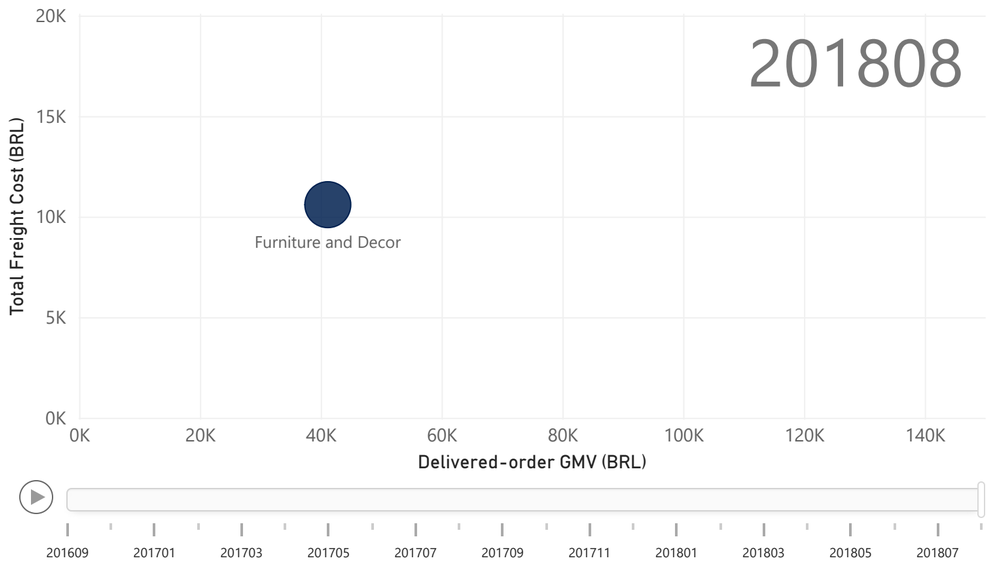
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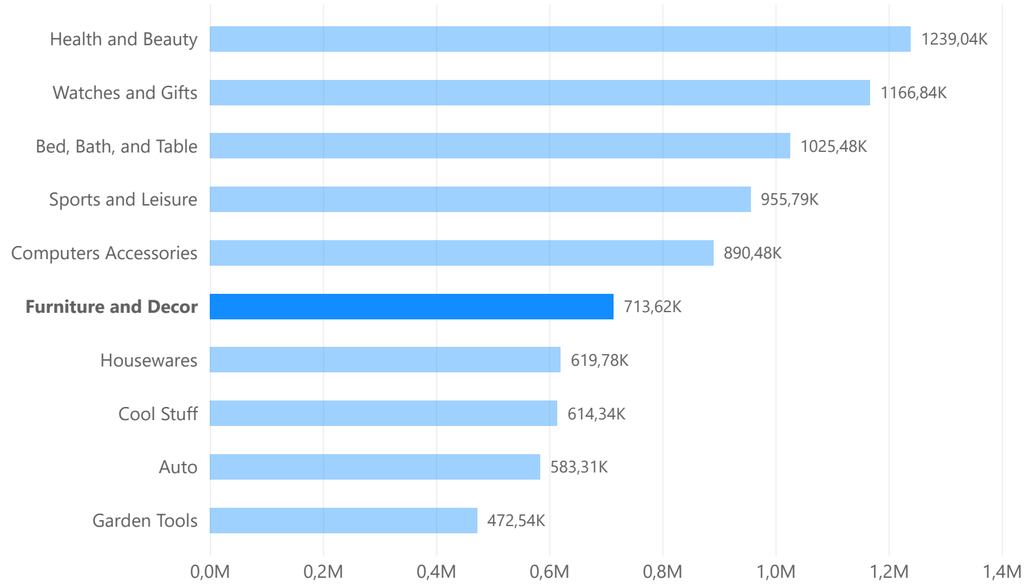
Selected period: 2016, 2017, 2018

5,4%
Category Share of GMV

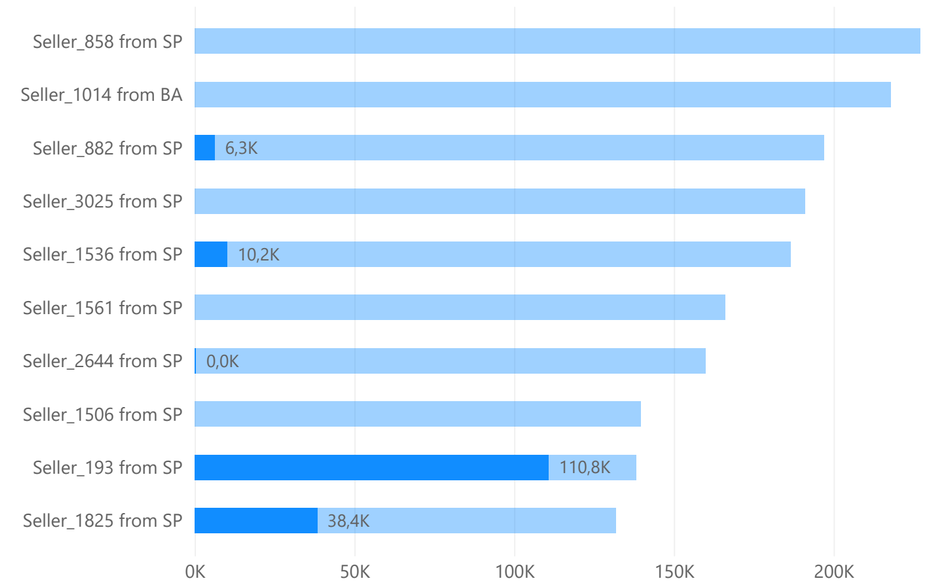
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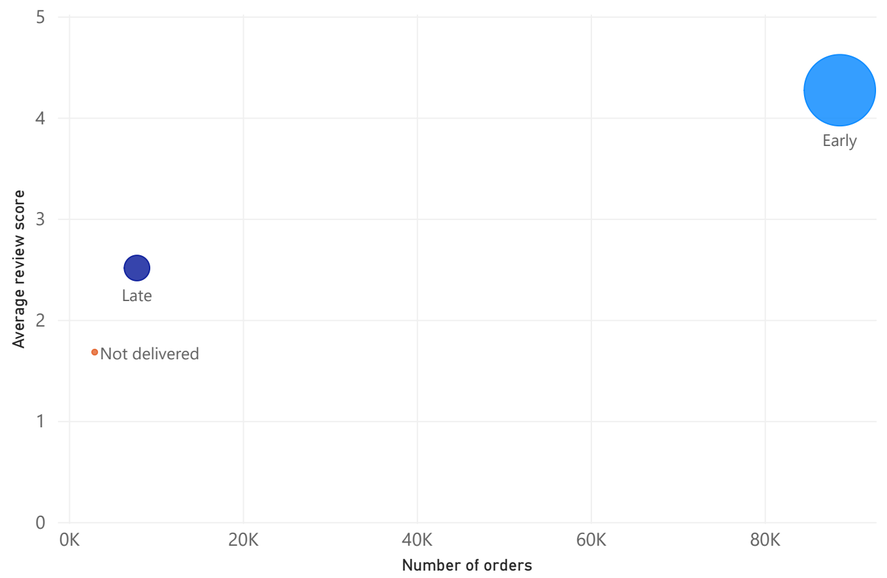
2016 | 2017 | 2018

Delivery Analysis

Selected period: 2016, 2017, 2018

4,05
Average review score

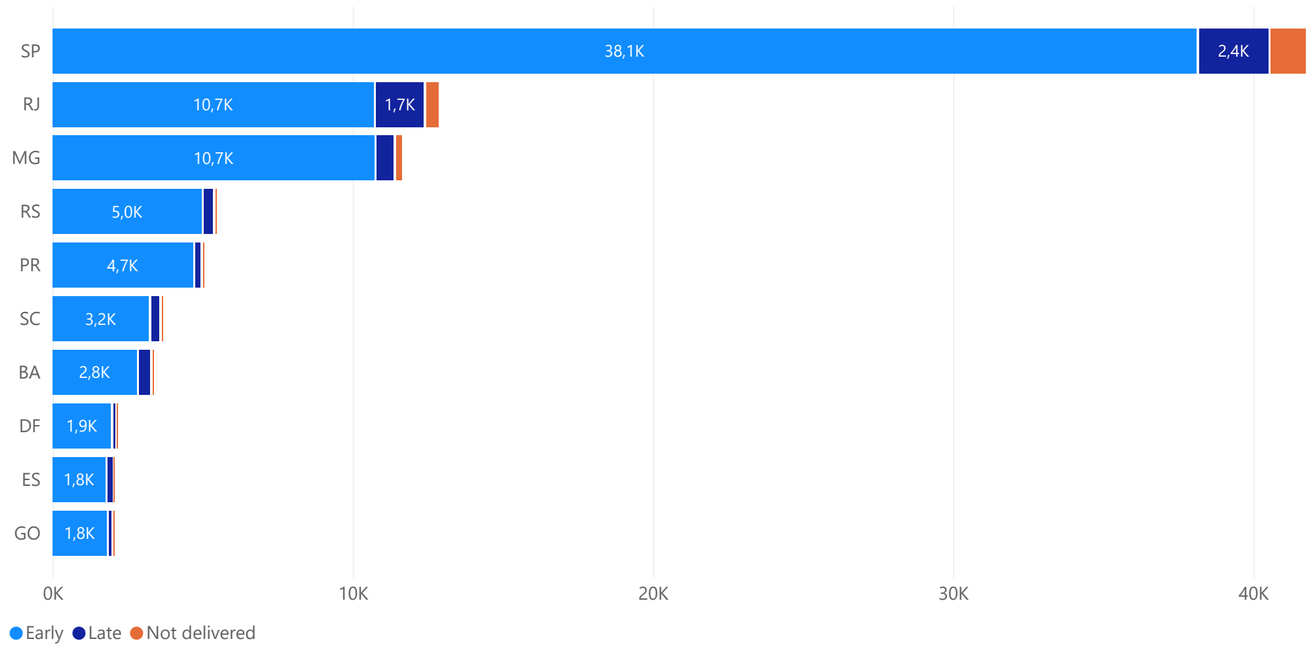
Delivery status of orders and average review score



Top 10 States with the highest late deliveries share

State	Number of late deliveries	Late deliveries share
AL	95	23,9%
MA	141	19,7%
PI	76	16,0%
CE	196	15,3%
SE	51	15,2%
BA	457	14,0%
RJ	1664	13,5%
TO	35	12,8%
PA	117	12,4%
ES	244	12,2%

Top 10 States by Delivered Orders



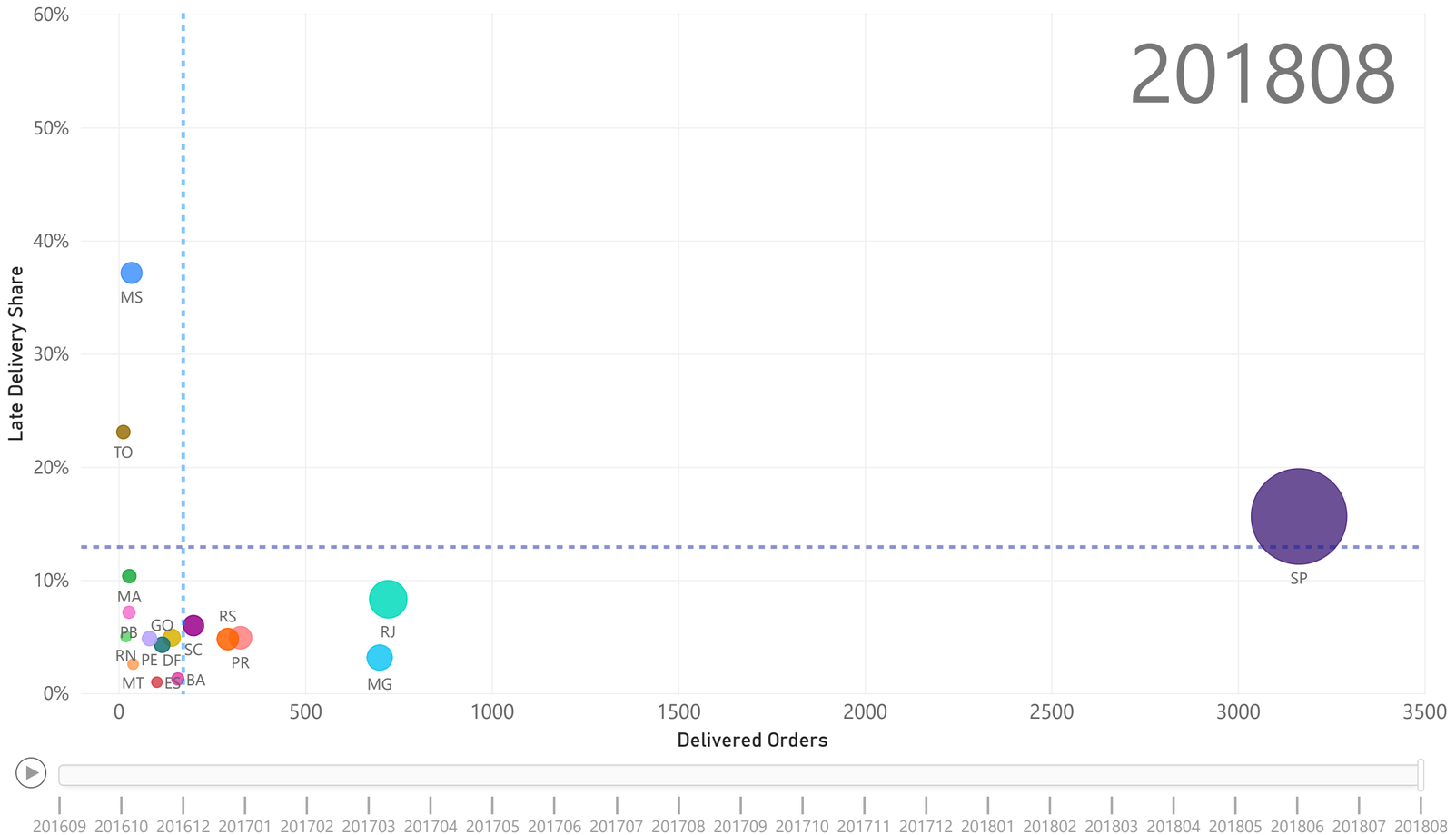
2016 | 2017 | 2018

Delivery Risk by State

Selected period: 2016, 2017, 2018

Delivered orders	Late Delivery Orders	Late Delivery Share	Average review score
96K	8K	8,1%	4,05

Delivered orders, late delivery share and late delivery orders by state



Late delivery share and average review score trends

